

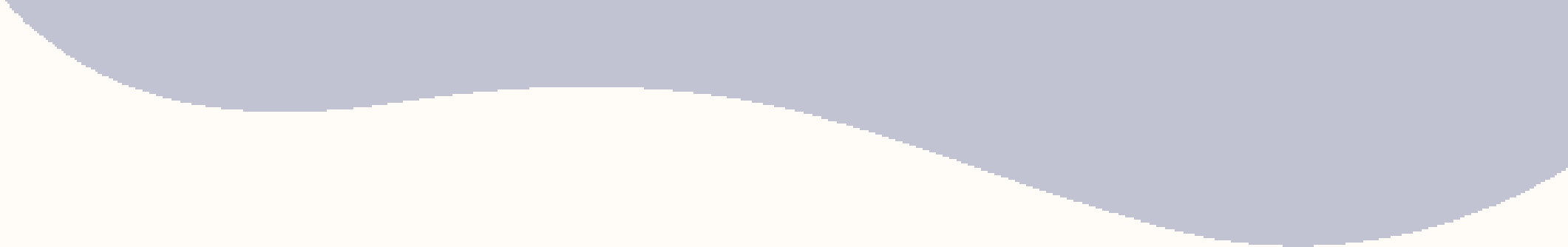
Road Accident Dashboard Analysis

Stakeholder Problem Statement

Transport authorities, police departments, and emergency response teams require a clear understanding of accident patterns across regions, vehicle types, road conditions, and severity levels.

They lack a centralized, data-driven view to identify high-risk roads, monitor yearly trends, and evaluate the causes of accidents across 2021 and 2022.

The purpose of this analysis is to provide actionable insights that improve road safety planning, resource allocation, and public awareness campaigns.



Insights

+

Recommendations

Accident Severity

- Slight injuries dominate, but 66K+ severe cases highlight critical safety gaps.
- Target high-severity zones for structural upgrades and stricter regulation.

Vehicle Type

- Cars are the largest contributor. Bikes and vans show similar risk levels.
- Awareness for car drivers & protective infrastructure for bikers.

Road Type

- Nearly 75% of casualties occur on single carriageways.
- Add dividers, widen lanes, and improve signage.

Monthly Trend

- 2022 consistently records lower casualties than 2021.
- Use seasonal trends for targeted awareness and enforcement.

Road Surface & Environment

- Most accidents occur on dry roads (normal conditions), while night-time crashes tend to be more severe.
- Improve lighting, add reflectors, strengthen winter safety measures.

Urban vs Rural

- Urban areas show higher accident frequency; rural areas often show higher severity.
- Urban traffic control + improved emergency access in rural zones.

Core Insights

- Cars & single carriageways are the highest-risk categories
- 2022 shows meaningful decline vs 2021
- Dry surfaces dominate incident counts
- Daylight hours show more accidents; nights show more severe incidents
- Urban accident volume > rural accident volume

Recommendation

- Improve single carriageway infrastructure
- Stricter speed enforcement
- Night-time visibility enhancements
- Seasonal (winter/monsoon) safety campaigns
- Faster rural emergency response
- Awareness programs for high-risk drivers

Executive Summary

Insights point toward focused upgrades in infrastructure, visibility, regulation, and seasonal safety initiatives.

- The dataset of 417,883 casualties reveals clear accident patterns across severity, road types, vehicle categories, and environmental conditions.
- Cars and single carriageways account for the majority of incidents.
- Urban regions record higher frequency, while rural areas show higher severity.
- Accident counts improve in 2022, indicating progress in safety measures.

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THANK YOU